

MAXWELL BURSEY

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EDUCATION

B.Eng. in Electrical Engineering (Co-op) — Term 5 of 8

2023 – 2028

Memorial University of Newfoundland — St. John's, NL

GPA: 3.87 / 4.0 (85.5%) Expected Graduation: May 2028

High School Diploma, Honours

Grad. 2022

Corner Brook Regional High School — Corner Brook, NL

Final Average: 95%

EXPERIENCE

Co-op Engineering Student

Aug. 2025 – Dec. 2025

The Commons, Queen Elizabeth II Library, Memorial University — St. John's, NL

- Independently scoped, designed, and executed a workstation usage-tracking evaluation project from inception to final report, demonstrating end-to-end engineering project ownership in a professional environment.
- Prototyped and benchmarked sensing approaches including PIR motion detection and AC current measurement; developed signal classification logic to reduce false positives and suppress transient noise under real-world conditions.
- Produced a formal technical report and supervisor presentation addressing data privacy constraints, long-term scalability, and maintenance requirements, enabling informed operational decision-making.
- Maintained weekly reporting cadence with supervisor, aligning engineering deliverables with institutional requirements and project timelines throughout the work term.
- Coordinated asynchronously with cross-functional colleagues via Slack and Trello, ensuring timely issue escalation and smooth collaborative workflows.

Head Server

Jul. 2022 – Aug. 2023

WOWOs Asian Homestyle Cooking — Corner Brook, NL

- Managed dining floor operations for a high-volume restaurant, maintaining consistently high guest satisfaction ratings across a full 13-month tenure.
- Recruited, onboarded, and trained new service staff; established team standards that reduced order errors and improved overall service consistency.
- Configured and commissioned new point-of-sale payment systems and digital menus, applying technical skills directly in a live operational environment.

Equipment Operator & Office Assistant

Jun. 2021 – Sep. 2021

Trademark Lumber Company — Deer Lake, NL

- Operated industrial lumber processing machinery under strict safety protocols, contributing to a zero-incident season across the full work term.
- Processed employee timecards and calculated weekly payroll with 100% accuracy, eliminating compensation errors for the complete workforce.
- Supported yard logistics and inventory tracking, coordinating seamlessly between outdoor operations and office administration.

PROJECTS

Analog Voltage Reference PCB Design

Jan. – Feb. 2026

Personal / Academic Engineering Project

- Designed and developed a two-layer PCB providing a stable, low-noise 5 V analog reference from a 3-cell LiPo battery, implementing reverse polarity protection, linear voltage regulation, and buffered output for precision analog applications.
- Created the complete schematic and PCB layout in KiCad, performing component selection, footprint assignment, custom battery connector footprint development, and routing with dedicated power nets and a continuous ground plane to improve signal integrity and reduce noise.
- Engineered the reference circuit using an NCP1117 linear regulator, zener reference network, and MCP6001 rail-to-rail op-amp configured as a voltage follower to produce a low-impedance, stable reference output accessible through test points and output connectors.
- Performed ERC/DRC verification and manufacturing preparation, generating Gerber files, drill files, a detailed bill of materials, and 3D PCB visualization to validate layout, manufacturability, and component placement.

Laptop Station Usage Tracking System

Oct. 2025 – Dec. 2025

Work Term Project – The Commons, Memorial University

- Designed and evaluated an automated sensing system to classify active vs. inactive workstation states in a shared academic computing lab, assessing accuracy, reliability, scalability, and operational constraints.
- Prototyped PIR motion-detection and AC current-sensing circuits; implemented software classification logic to minimize false positives and suppress transient noise under real-world conditions.
- Delivered findings via a formal technical report and presentation, addressing data privacy constraints, long-term scalability, and maintenance requirements for supervisor decision-making.

Automatic Irrigation System

Feb. 2024 – Apr. 2024

Personal Project – St. John's, NL

- Designed and built a fully closed-loop automated irrigation system integrating a microcontroller, capacitive soil moisture sensors, relay-driven water pump, LCD status display, and indicator LEDs.
- Programmed threshold-based control logic in C; calibrated sensors for repeatable, accurate readings under varying soil conditions, ensuring consistent and reliable autonomous operation.

Password Generation Program

Sep. 2023 – Dec. 2023

Introduction to Programming – Memorial University

- Developed a Python CLI application generating policy-compliant passwords by combining user-defined constraints, character class rules, and cryptographic randomization via the `secrets` module.
- Validated program correctness through systematic test cases; presented design methodology and outcomes to a class audience with a structured technical presentation.

SKILLS & CERTIFICATIONS

Programming Python | C | C++

Engineering Software KiCad | MATLAB | SolidWorks | Onshape | PSpice

Productivity & Collab. Microsoft Office 365 (Word, Excel, PowerPoint) | Slack | Trello

Core Competencies PCB design & layout | Embedded systems | Sensor integration | Technical documentation | Engineering analysis

Certifications WHMIS | Tech Stewardship Micro-Credential | Microsoft Excel Essential Training | Full Class 5 Driver's Licence